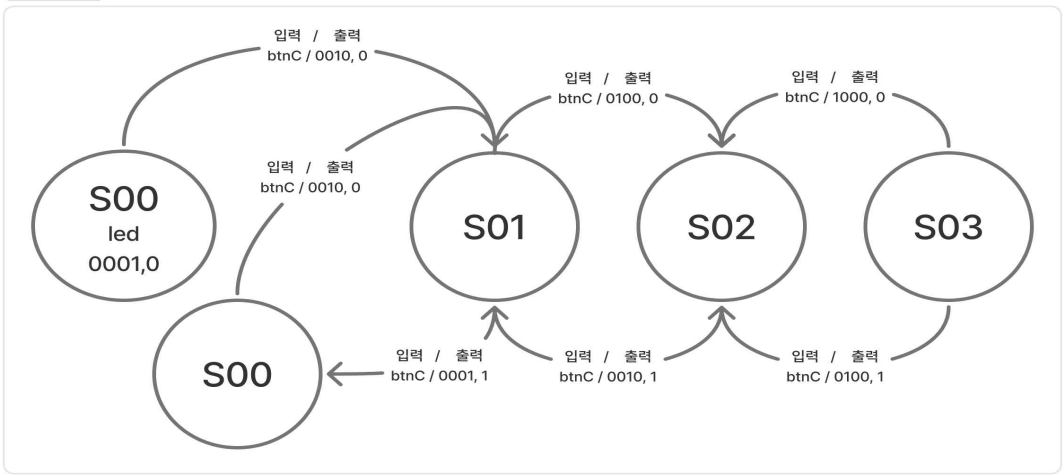
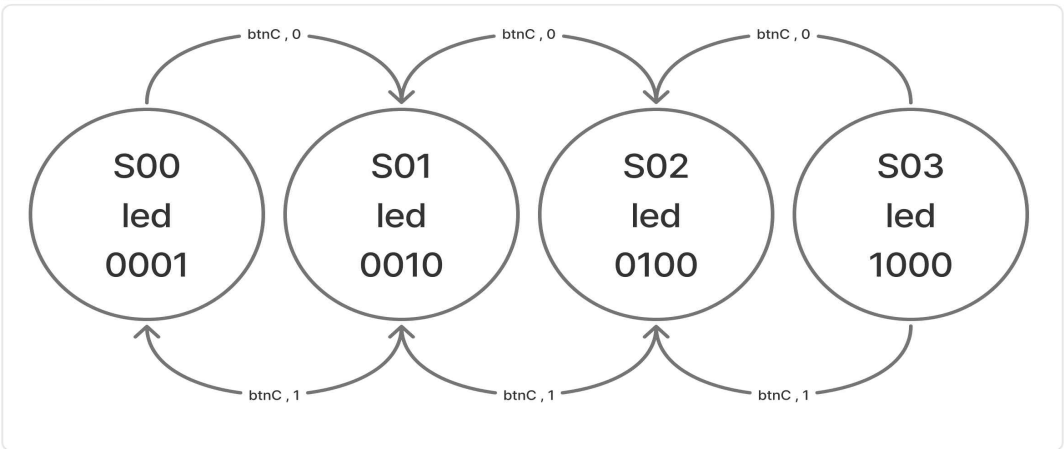
led fsm melay machine 구현

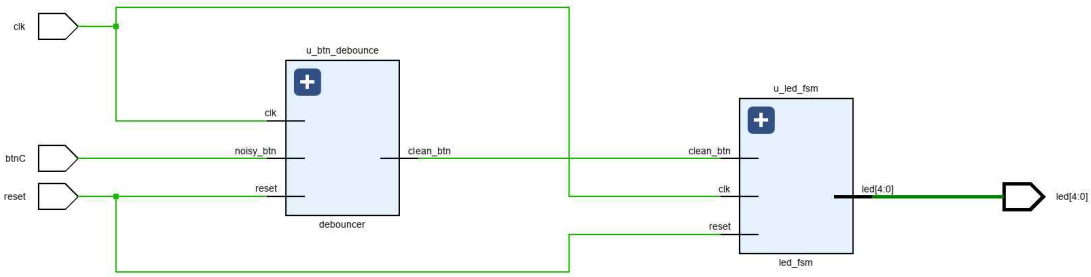
led_fsm function block diagram

Mealy_fsm



Moore_fsm





led_fsm source code

led_fsm.v

```
module led_fsm(
    input clk,
    input reset,
    input clean_btn,
    output reg [4:0] led
);

    localparam [1:0] S0 = 2'b00,
                     S1 = 2'b01,
                     S2 = 2'b10,
                     S3 = 2'b11;

    reg [1:0] state, next_state;
    reg dir, next_dir;
    reg prev_btn;

    wire btn_pulse;

    // 버튼을 누른 순간만 1
    assign btn_pulse = clean_btn & ~prev_btn;

    // 이전 버튼 상태 저장
    always @(posedge clk or posedge reset) begin
        if (reset)
            prev_btn <= 0;
        else
            prev_btn <= clean_btn;
    end

    // 현재 상태 및 방향 저장
    always @(posedge clk or posedge reset) begin
        if (reset) begin
            state <= S0;
            dir <= 0;
        end
        else begin
```

```
            state <= next_state;
            dir <= next_dir;
        end
    end

    // 다음 상태 결정
    always @(*) begin
        next_state = state;
        next_dir = dir;

    if (btn_pulse) begin
        if (dir == 0) begin  // 정방향
            if (state == S3) begin
                next_state = S2;
                next_dir = 1;
            end
            else
                next_state = state + 1'b1;
        end
        else begin           // 역방향
            if (state == S0) begin
                next_state = S1;
                next_dir = 0;
            end
            else
                next_state = state - 1'b1;
        end
    end
end
    // Mealy 출력
    always @(*) begin
        led = 5'b00000;

        led[3:0] = 4'b0001 << next_state;

        // 역방향이면 LED4 ON
        led[4] = next_dir;
    end
endmodule
```

led_fsm test bench

tb_led_fsm.v

```
`timescale 1ns / 1ps
module tb_led_fsm();

    reg clk = 0;
    reg reset = 0;
    reg clean_btn = 0;

    wire [4:0] led;

    reg r_error = 1'b0;

    // debounce 제외하고 FSM만 테스트
    led_fsm u_led_fsm(
        .clk(clk),
        .reset(reset),
        .clean_btn(clean_btn),
        .led(led)
    );

    always #5 clk = ~clk; // 100MHz clk 생성

    // 버튼을 한 번 누르고 LED 결과 확인
    task send_data(input [4:0] expect_val);
        begin
            @(negedge clk);
            clean_btn = 1;

            @(posedge clk); #1;

            if (led !== expect_val) begin
                r_error = 1'b1;
                    $display("FAIL  t=%0t  led=%b  expect=%b", $time, led,
expect_val);
            end
```

```
            @(negedge clk);
            clean_btn = 0;

            @(posedge clk); #1;
        end
    endtask
    initial begin
        #100;
        reset = 1;
        #100;
        reset = 0;

        // 초기 LED0
        #1;
        if (led !== 5'b00001)
            r_error = 1'b1;

        // 정방향
        send_data(5'b00010); // LED1
        send_data(5'b00100); // LED2
        send_data(5'b01000); // LED3

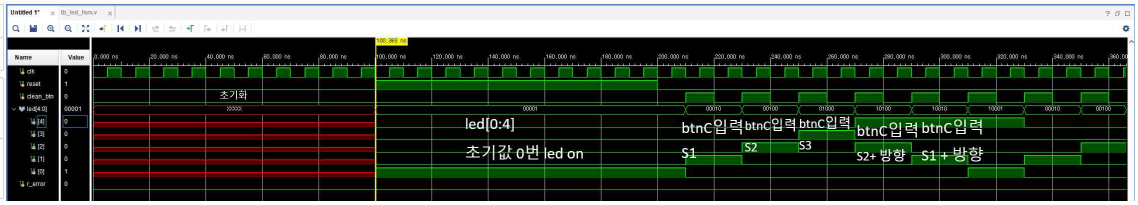
        // 역방향 (LED4 ON)
        send_data(5'b10100); // LED4 + LED2
        send_data(5'b10010); // LED4 + LED1
        send_data(5'b10001); // LED4 + LED0

        // 다시 정방향
        send_data(5'b00010); // LED1
        send_data(5'b00100); // LED2

        if (r_error == 0)
            $display("PASS melay FSM");
        else
            $display("FAIL melay FSM");

        $finish;
    end
endmodule
```

test bench 결과




```
# run 1000ns
PASS melay FSM
$finish called at time : 366 ns : File "D:/Project/fpga/ai2/06_1.btn_debounce_10ms/06_1.btn_debounce_10ms.srcs/sim_1/new/tb_led_fsm.v" Line 73
```